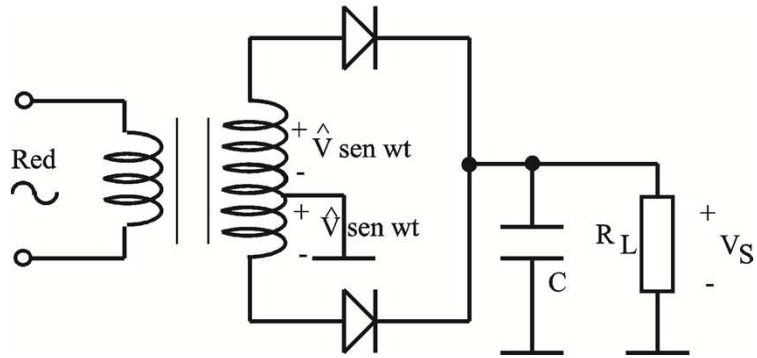
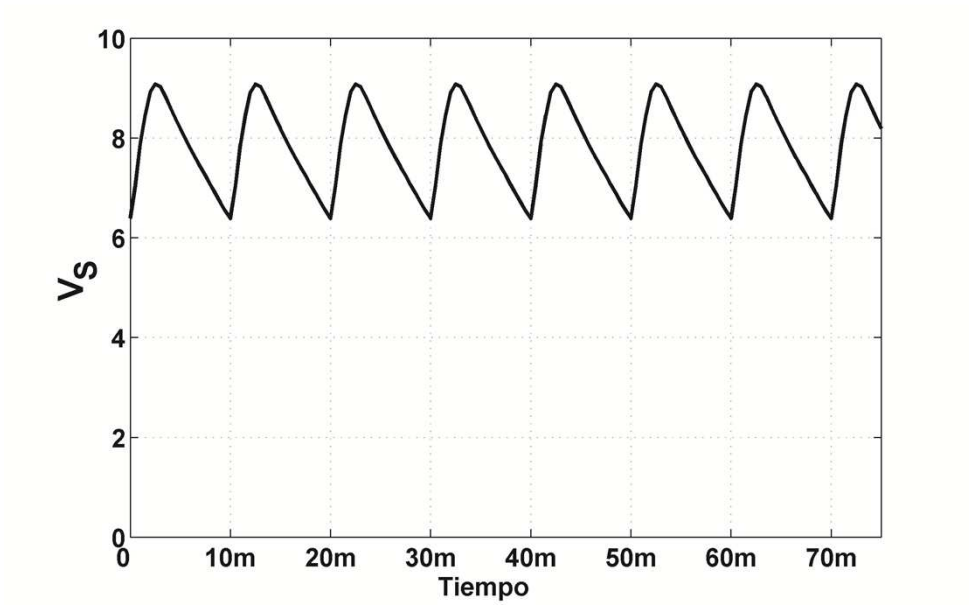
Fuentes no Reguladas

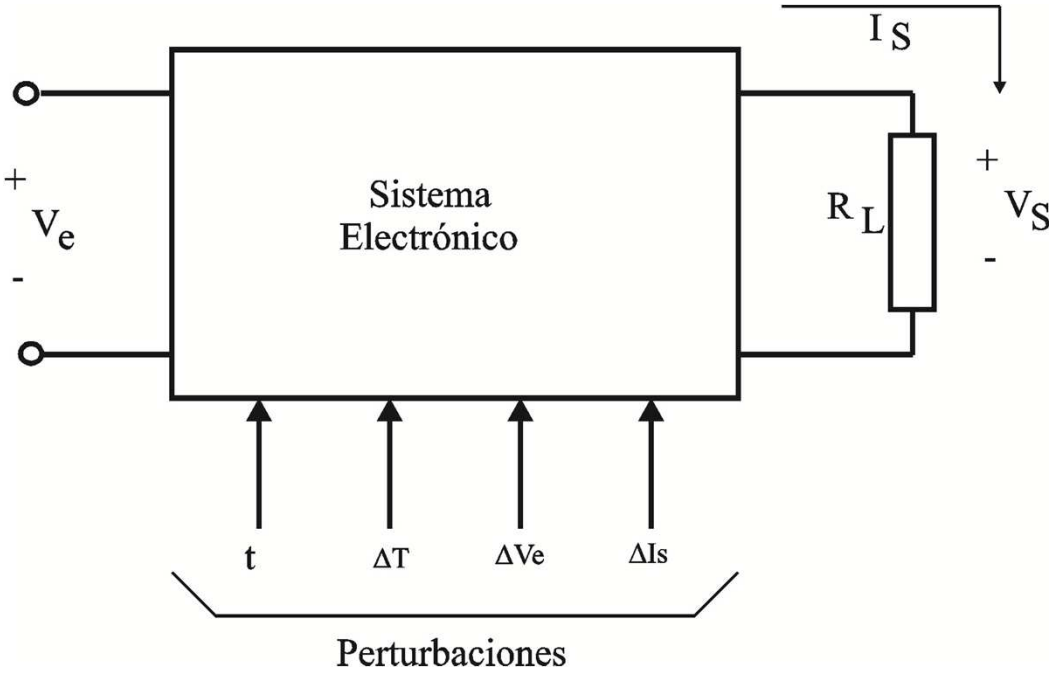




Fuentes no Reguladas

- ✓ La tensión de la red puede tener variaciones importantes.
- ✓ El circuito de carga (simulado por RL) puede absorber más o menos corriente, por lo que las caídas de tensión en el transformador y en los diodos producirán variaciones apreciables en el valor medio de la tensión de salida.
- ✓ El circuito rectificador deja pasar componentes alternas de rizado, también dependientes de la carga.

Fuentes Reguladas



El sistema es una fuente de tensión regulada, si teniendo en cuenta las posibles perturbaciones:

$$\frac{\Delta V_s}{V_s} < \frac{\Delta V_e}{V_e}$$

Factor de Regulación

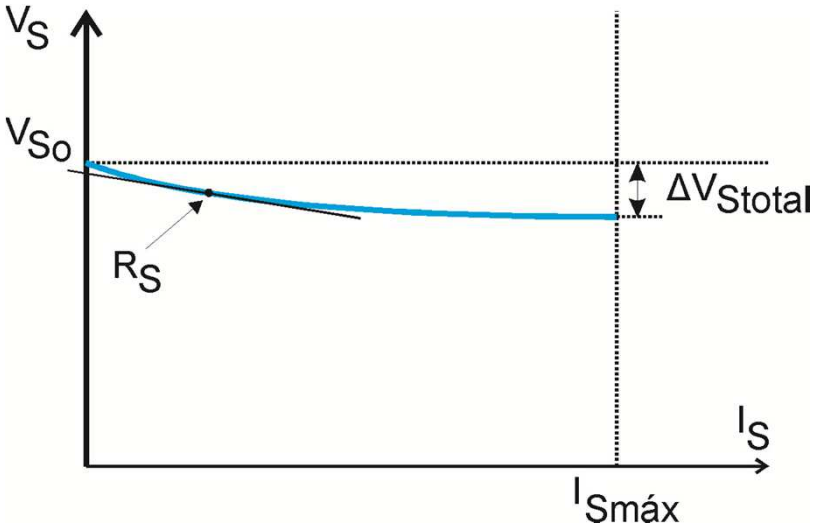
$$F_R = \left.\frac{\Delta V_s}{\Delta V_e}\right|_{\Delta T=0\,,\,\Delta I_s=0}$$

- ✓ Factor de regulación de línea. Variación del valor medio de V_e
- ✓ Factor de regulación del ripple (rizado).

$$F_R{}'\% = \frac{\frac{\Delta V_s}{V_s}}{\frac{\Delta V_e}{V_e}} 100\%$$

- ✓ Fuentes de gran calidad deberán tener un $F_R{}'\% < 0{,}01\%$
- ✓ Fuentes de calidad media: $F_R{}'\% \approx 0{,}1\%$

Resistencia de Salida



$$R_S = \left.\frac{\Delta V_S}{\Delta I_S}\right|_{\Delta V_e=0\ ,\ \Delta T=0}$$

$$|Z_S| = \left|\frac{\Delta V_S}{\Delta I_S}\right| (w)$$

$$R_{Stotal} = \frac{\Delta V_{Stotal}}{I_{Smáx}}$$

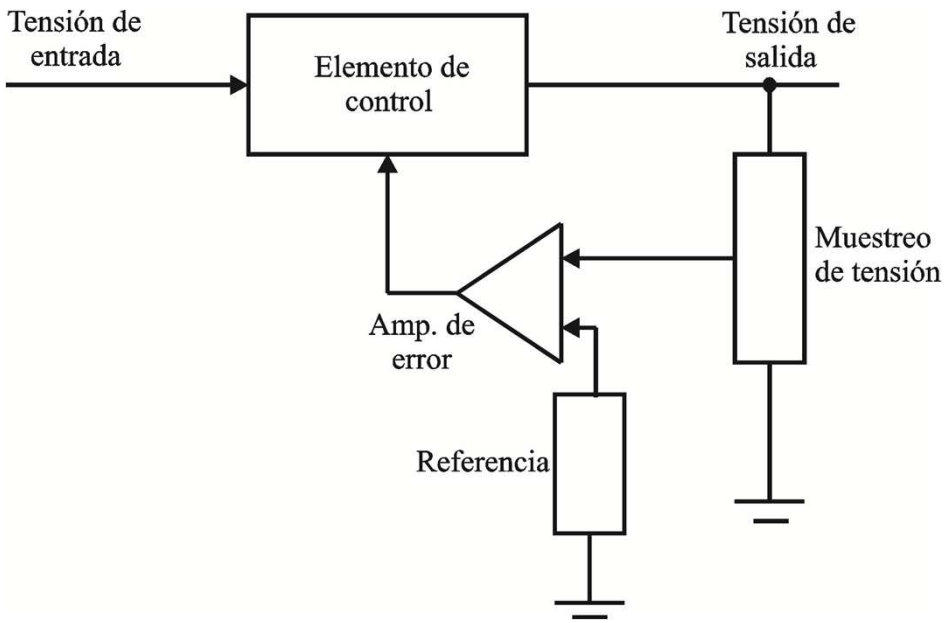
Coeficiente de temperatura

Para estudiar el comportamiento con respecto a la variación de temperatura se define el coeficiente de temperatura como:

$$K_T = \left.\frac{\Delta V_S}{\Delta T}\right|_{\Delta V_e=0\ ,\ \Delta I_S=0}$$

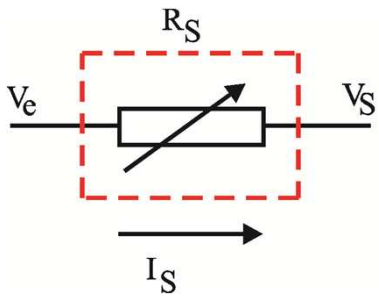
$$K_T\% = \frac{\frac{\Delta V_S}{V_S}}{\Delta T} 100\%$$

Regulador de tensión genérico

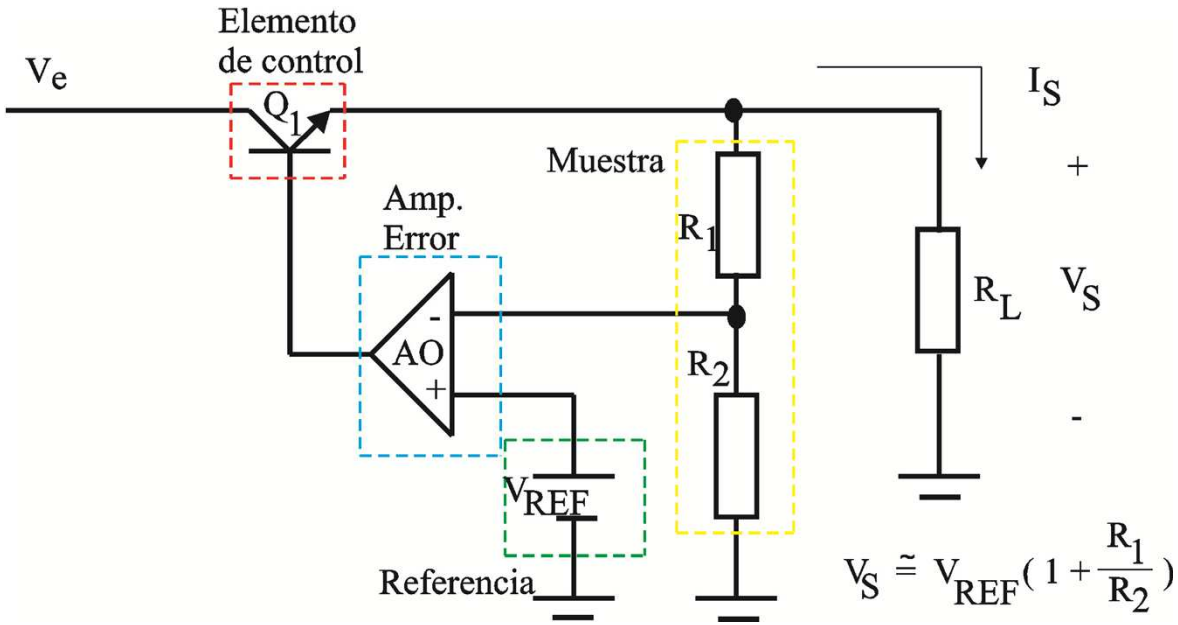


- ✓ Referencia de tensión
- ✓ Muestreo de tensión
- ✓ Amplificador de error
- ✓ Elemento de control

Regulador serie

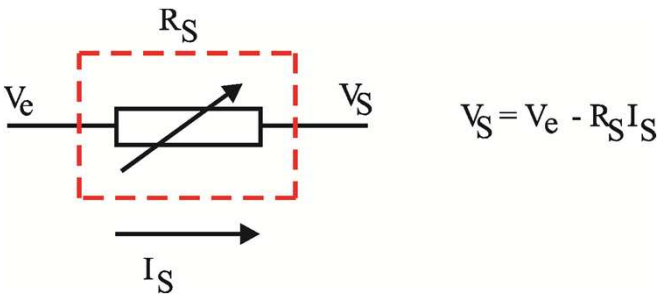


$$V_S = V_e - R_S I_S$$

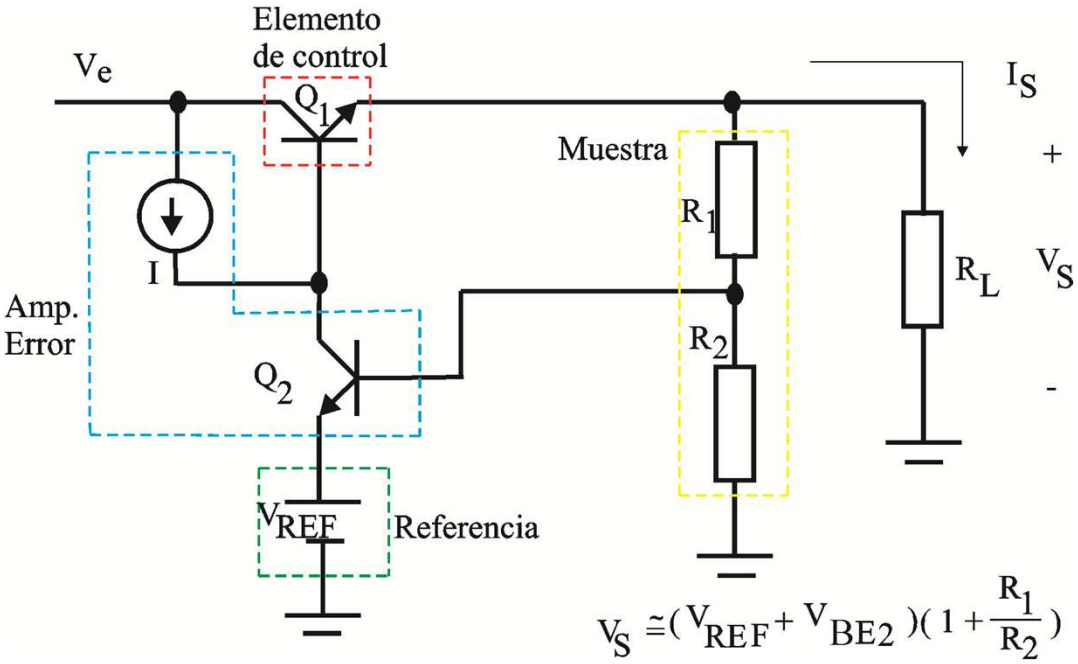


$$V_S \cong V_{REF}\left(1 + \frac{R_1}{R_2}\right)$$

Regulador serie

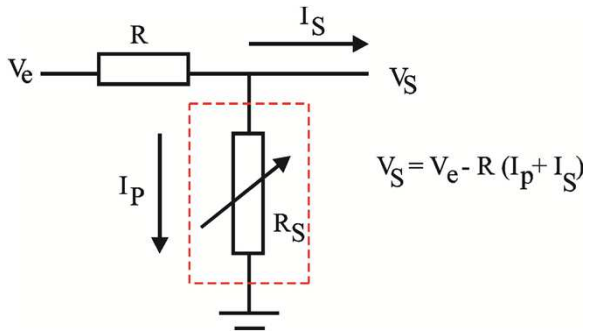


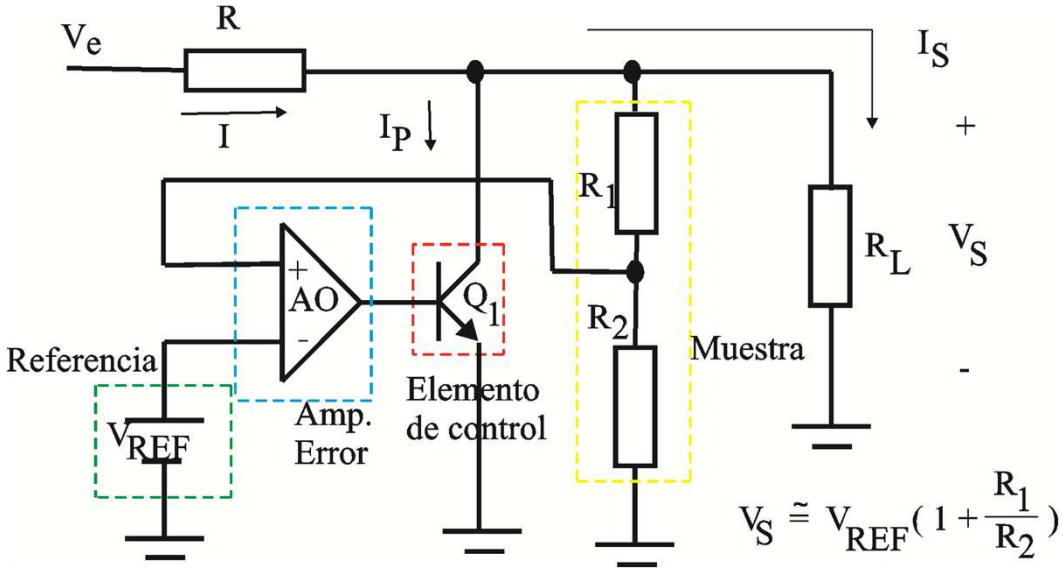
$$V_S = V_e - R_S I_S$$



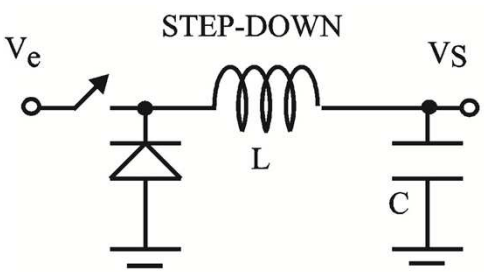
$$V_S \cong (V_{REF} + V_{BE2})\left(1 + \frac{R_1}{R_2}\right)$$

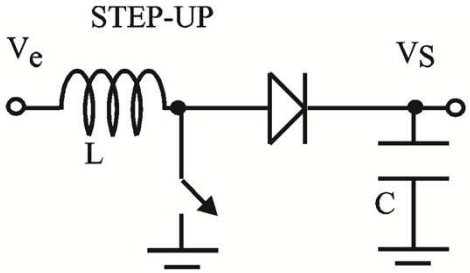
Regulador paralelo





Reguladores conmutados





ton tiempo en que la llave está cerrada
toff tiempo en que la llave está abierta

$$V_S = V_e \frac{ton}{ton + toff}$$

$$V_S = V_e \frac{ton + toff}{toff}$$

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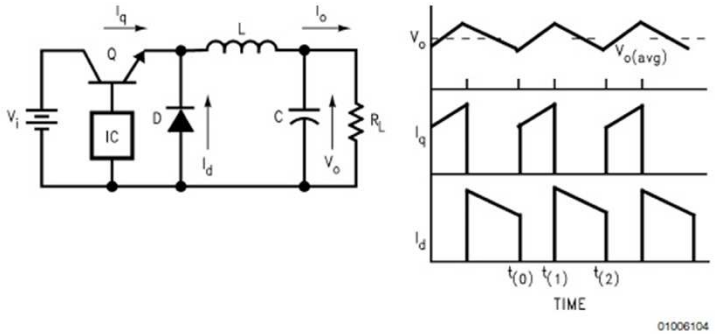


FIGURE 4. Buck Regulator Circuit with Voltage and Current Waveforms

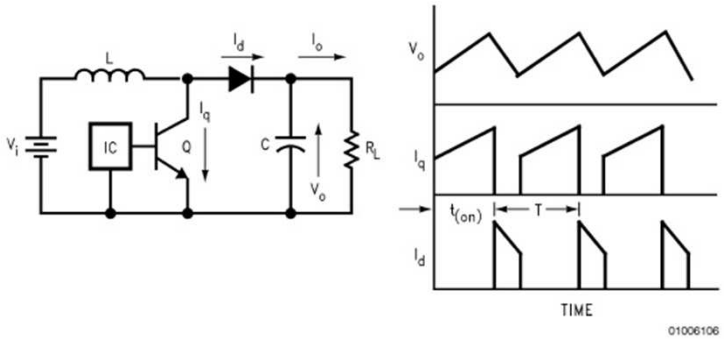
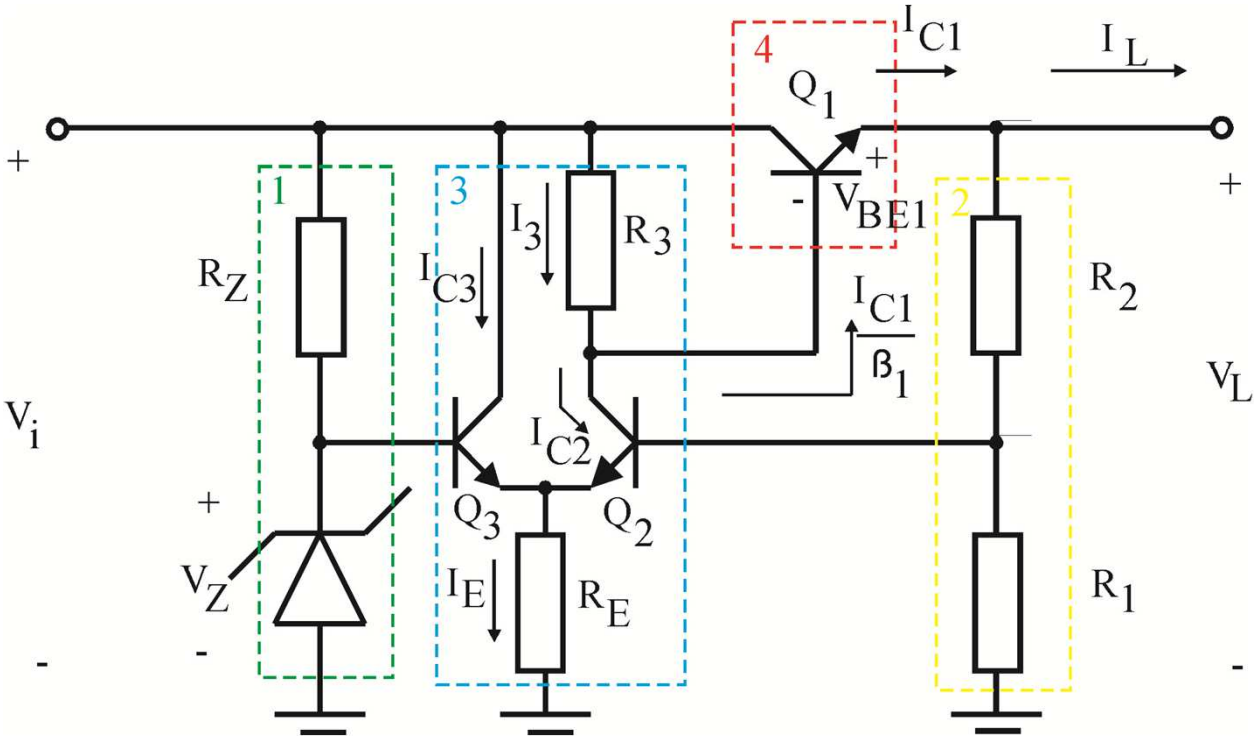


FIGURE 5. Boost Regulator and Associated I/V Waveforms

Ejemplo: Fuente regulada serie



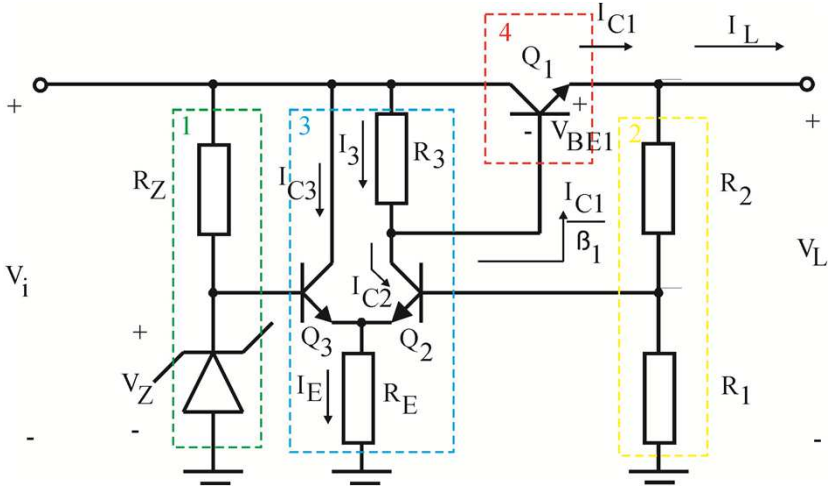
1. Referencia de tensión
2. Muestreo de tensión.
3. Amplificador de error
4. Elemento de control.

Ejemplo: Fuente regulada serie

Q_1 debe dimensionarse para:

$$P_{Q_1} = I_{Lmáx}(V_i - V_L)$$

$$I_3 = \frac{V_i - V_{BE1} - V_L}{R_3} = \frac{I_{C1}}{\beta_1 + 1} + I_{C2}$$



A máxima carga $I_{C1} \cong I_{Lmáx}$, R_3 debe diseñarse para proveer la corriente $I_3 > \frac{I_{Lmáx}}{\beta_1+1} + I_{C2mín}$, por lo tanto:

$$R_3 < \frac{V_{imín} - V_{BE1}(I_{Lmáx}) - V_L}{\frac{I_{Lmáx}}{\beta_1 + 1} + I_{C2mín}}$$

Ejemplo: Fuente regulada serie

Cuando $I_L = 0$, I_{C2} será máxima

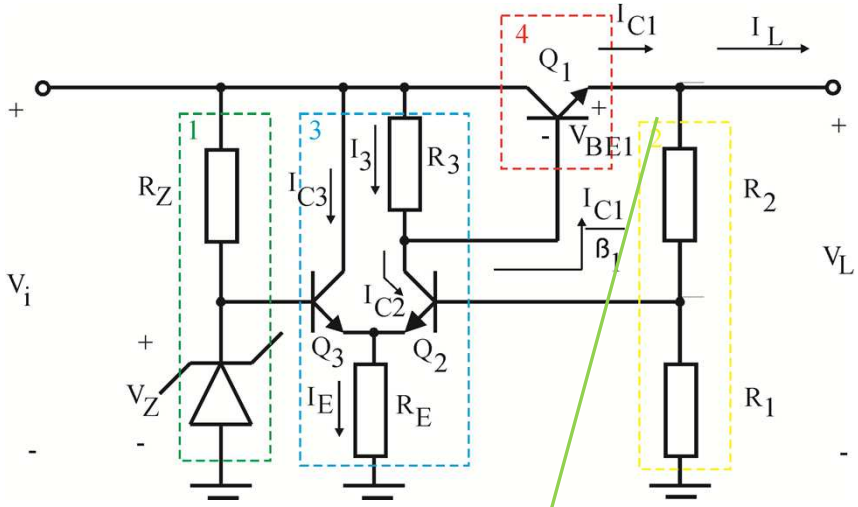
$$I_E = \frac{V_Z - V_{BE3}}{R_E} \cong I_{C2} + I_{C3}$$

$$I_{C2máx} \cong \frac{V_{imáx} - V_{BE1on} - V_L}{R_3}$$

Se diseña:

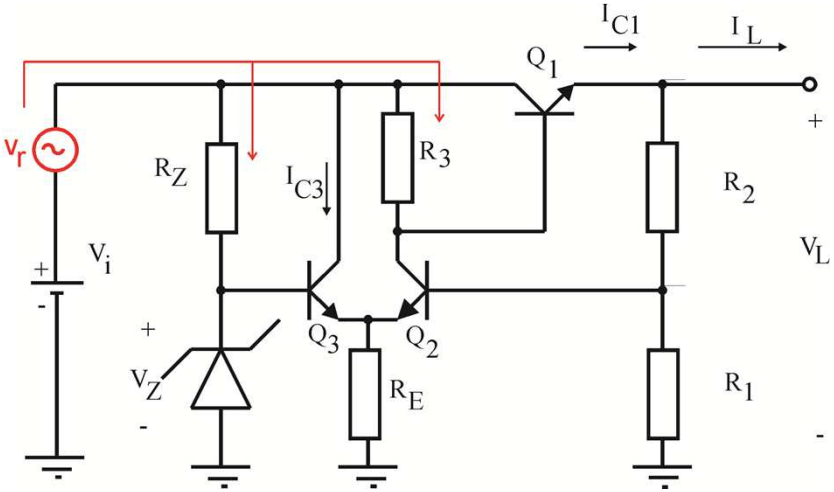
$$I_E > I_{C2máx} + I_{C3mín}$$

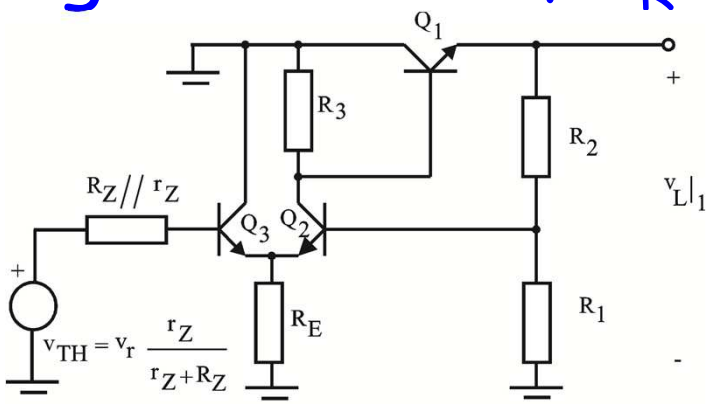
$$R_E < \frac{V_Z - V_{BE3}}{I_{C2máx} + I_{C3m'ln}}$$



$$V_L \cong V_Z \frac{R_1 + R_2}{R_1}$$

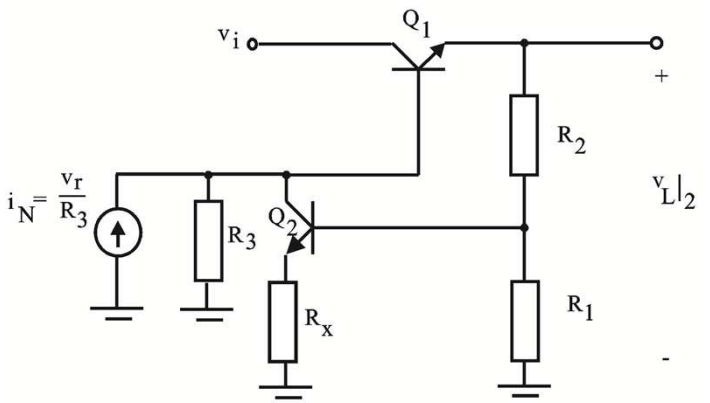
Ejemplo: Fuente regulada serie, F_R





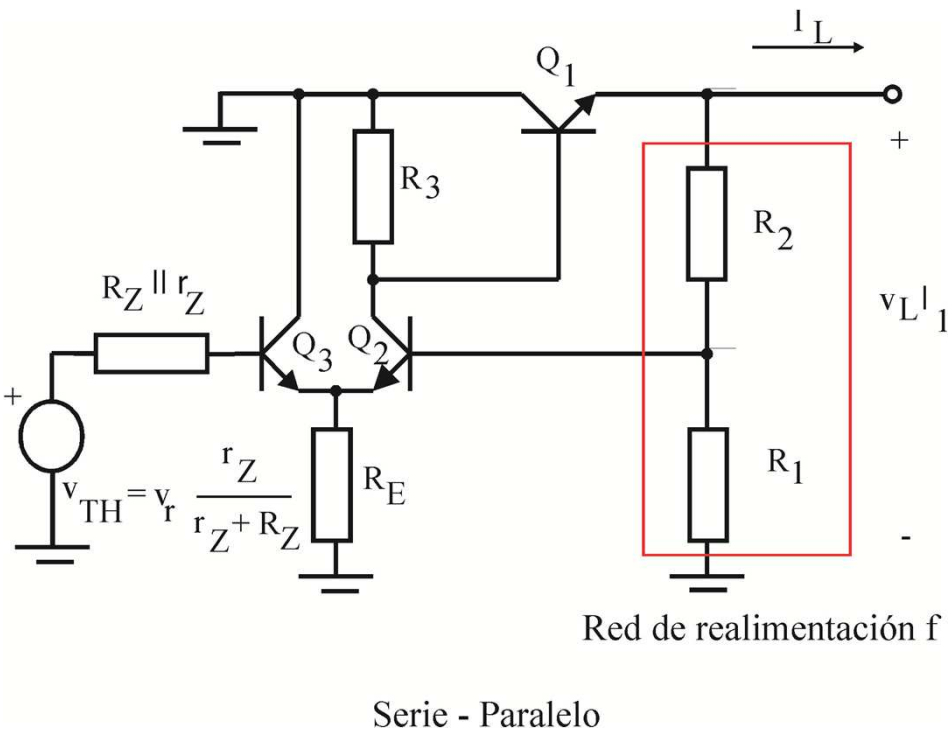
$$v_L = v_L\Big|_1 + v_L\Big|_2$$

$$F_R = \frac{v_L}{v_r}$$



$$R_x = R_E \| \frac{(R_Z \| r_Z) + h_{ie3}}{(h_{fe3} + 1)}$$

Ejemplo: Fuente regulada serie, F_R



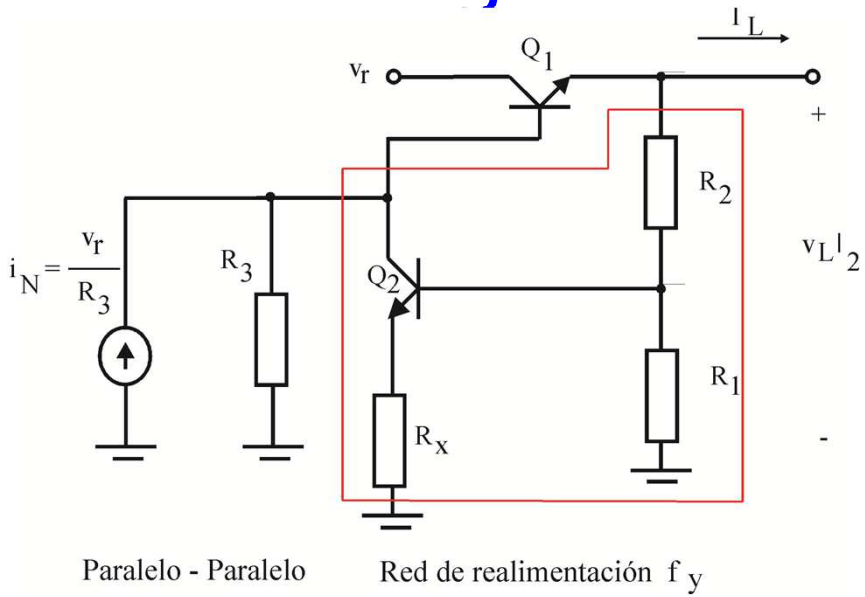
$$f = h_{12} = \frac{R_1}{R_1 + R_2}$$

$$Si\ \ T = af \gg 1 \rightarrow A \cong \frac{1}{f}$$

$$A = \frac{v_L|_1}{v_{TH}} \cong \frac{R_1 + R_2}{R_1}$$

$$\boxed{\frac{v_L|_1}{v_r} = \frac{v_L|_1}{v_{TH}}\frac{v_{TH}}{v_r} = \frac{R_1 + R_2}{R_1}\frac{r_z}{R_z + r_z}}$$

Ejemplo: Fuente regulada serie, F_R



$$f_y = y_{12} = \frac{h_{fe2}}{h_{ie2} + R_x(h_{fe2}+1)} \frac{R_1 \| (h_{ie2} + R_x(h_{fe2}+1))}{(R_1 \| (h_{ie2} + R_x(h_{fe2}+1)) + R_2)}$$

$$Si \;\; T = a_z f_y \gg 1 \rightarrow A_z = \frac{v_L|_2}{i_N} \cong \frac{1}{f_y} \quad ,$$

$$\boxed{\frac{v_L|_2}{v_r} = \frac{v_L|_2}{i_N}\frac{i_N}{v_r} = \frac{1}{f_y R_3}}$$

Ejemplo: Fuente regulada serie, F_R

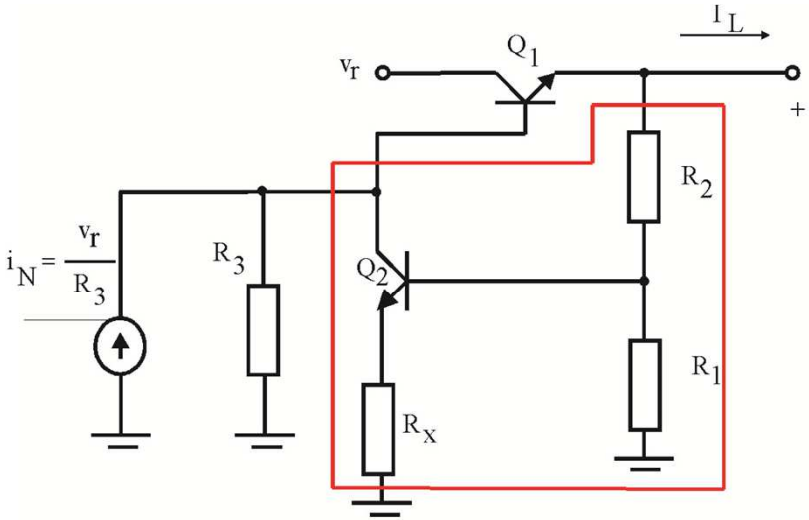
$$\frac{v_L|_1}{v_r} = \frac{v_L|_1}{v_{TH}}\frac{v_{TH}}{v_r} = \frac{R_1 + R_2}{R_1}\frac{r_z}{R_z + r_z}$$

$$\frac{v_L|_2}{v_r} = \frac{v_L|_2}{i_N}\frac{i_N}{v_r} = \frac{1}{f_y R_3}$$

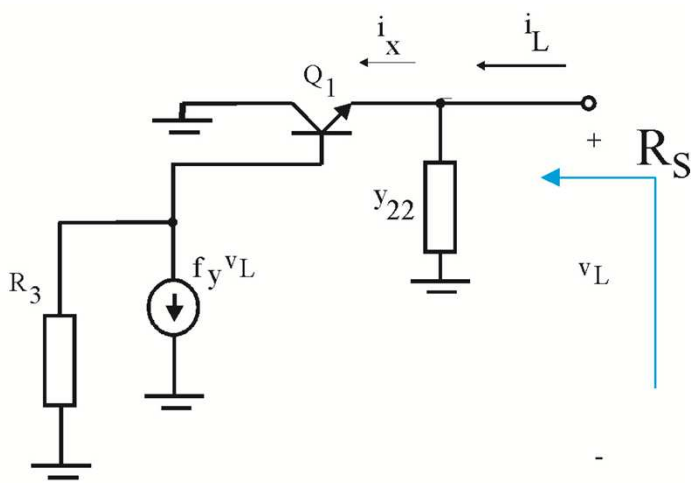
$$v_L = v_L\Big|_1 + v_L\Big|_2$$

$$F_R = \frac{v_L}{v_r} = \frac{R_1 + R_2}{R_1}\frac{r_z}{R_z + r_z} + \frac{1}{f_y R_3}$$

Ejemplo: Fuente regulada serie, R_S



Paralelo - Paralelo Red de realimentación f_y

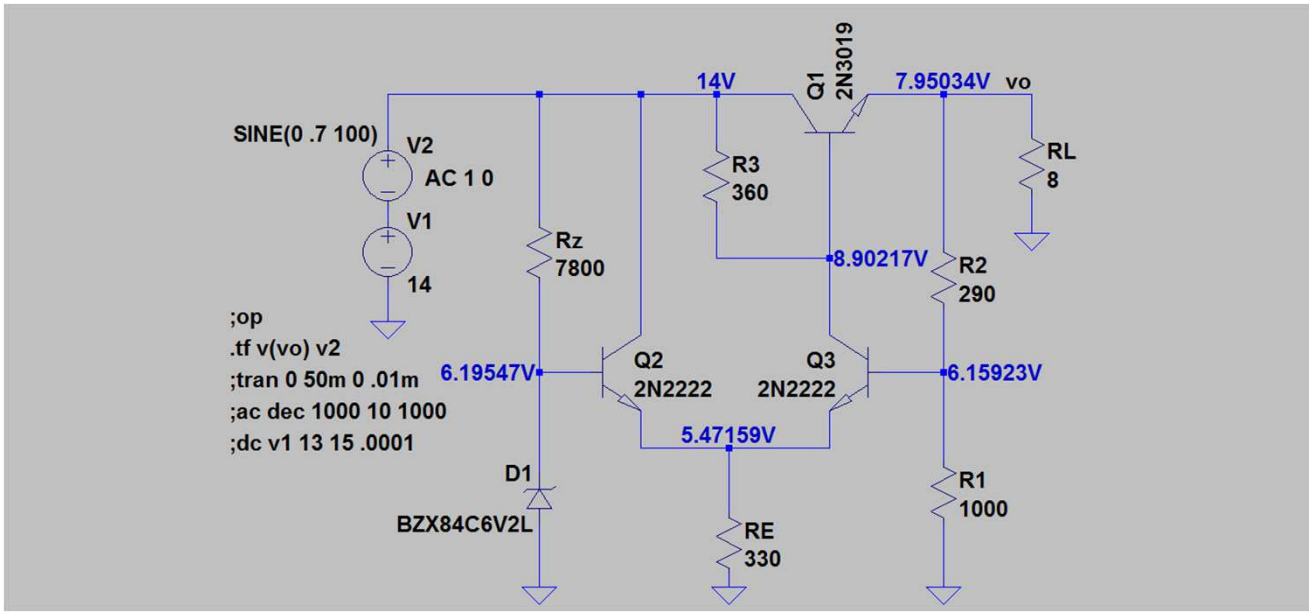


$$y_{22} = \frac{1}{R_2 + R_1 \| \left(h_{ie2} + R_x\left(h_{fe2} + 1\right)\right)} \quad , \qquad v_L = \frac{i_x}{h_{fe1} + 1} h_{ie1} + R_3 \left(\frac{i_x}{h_{fe1} + 1} - f_y v_L \right)$$

$$\frac{v_L}{i_x} = \frac{\dfrac{h_{ie1} + R_3}{h_{fe1} + 1}}{1 + R_3 f_y}$$

$$\boxed{R_s = \frac{1}{y_{22}} \Bigg\| \frac{v_L}{i_x}}$$

Ejemplo: Fuente regulada, simulación



Por cálculo, utilizando los parámetros obtenidos por simulación:

$F_R = 0{,}053$

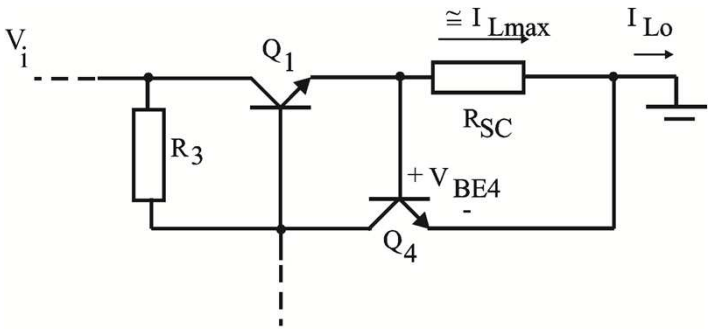
$R_s = 0{,}23\ \Omega$

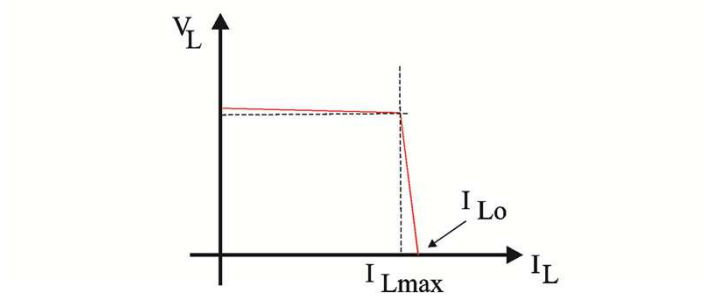
	Q1	Q2	Q3
hie	2,42	411	1520
hfe	59,3	199	201

rz	82,8

--- Transfer Function ---
Transfer_function:
0.0539003
output_impedance_at_V(vo):
0.240872

Protección contra sobrecargas y CC



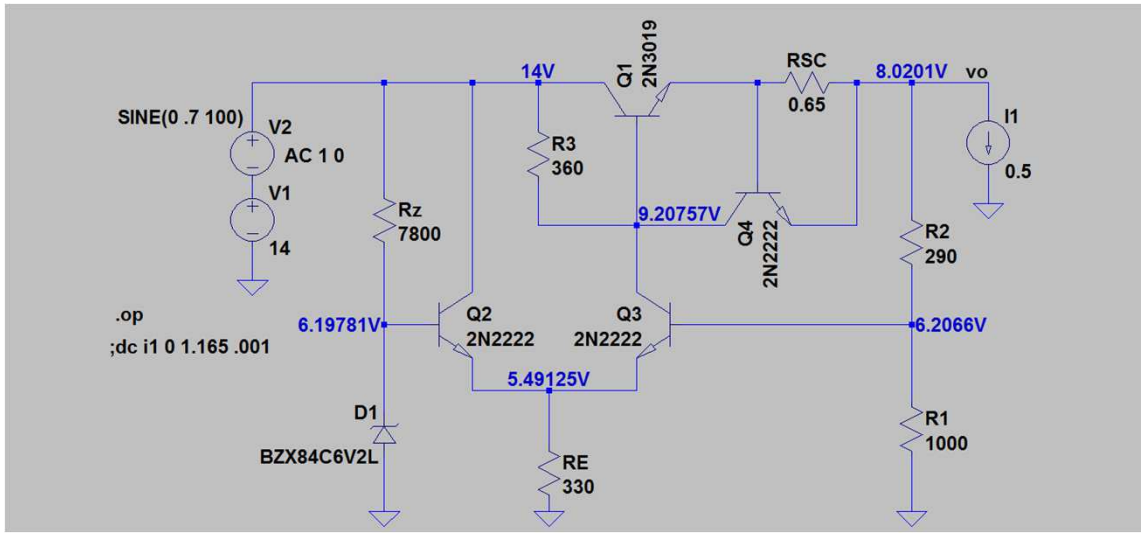


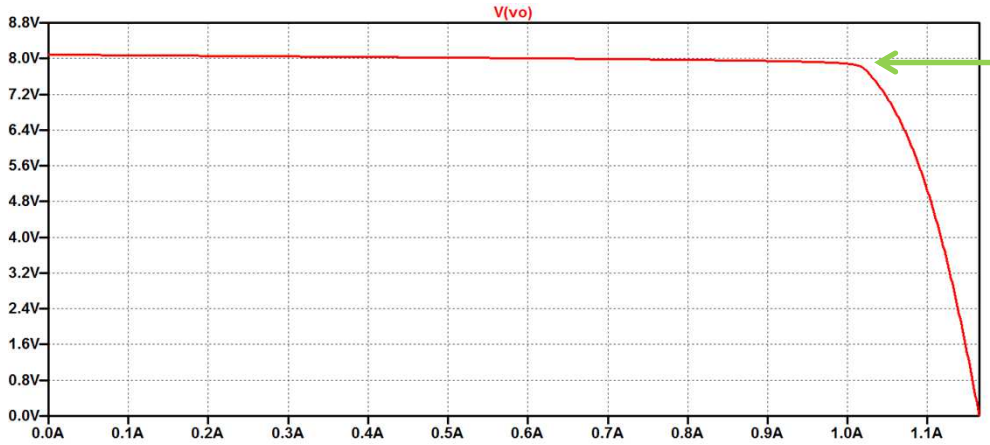
$$R_{SC} = \frac{V_{BE4}}{I_{Lmáx}}$$

$$I_{L0} = I_{Lmáx} + I_{C4}$$

$$P_{Q1} \cong V_i I_{L0}$$

Protección contra sobrecargas y CC

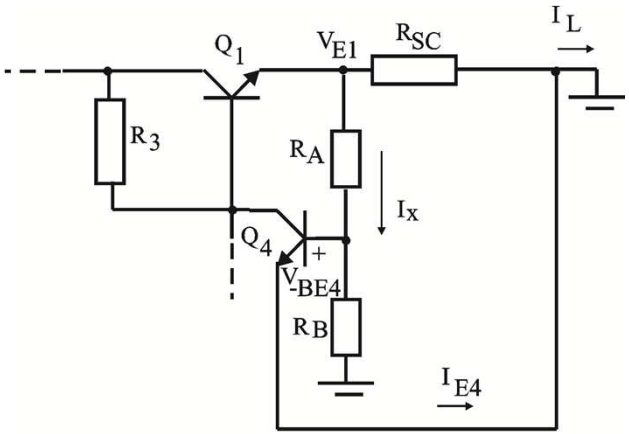


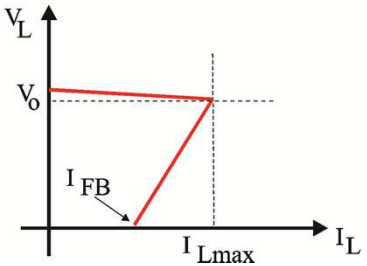


$I_{Lmáx}$

I_{L0}

Protección contra CC con repliegue (fold back)





Si $I_L = I_{lmáx}$:

$$\frac{R_B}{R_A + R_B}(I_{Lmáx}R_{SC} + V_L) - V_L = V_{BEon} \qquad (1)$$

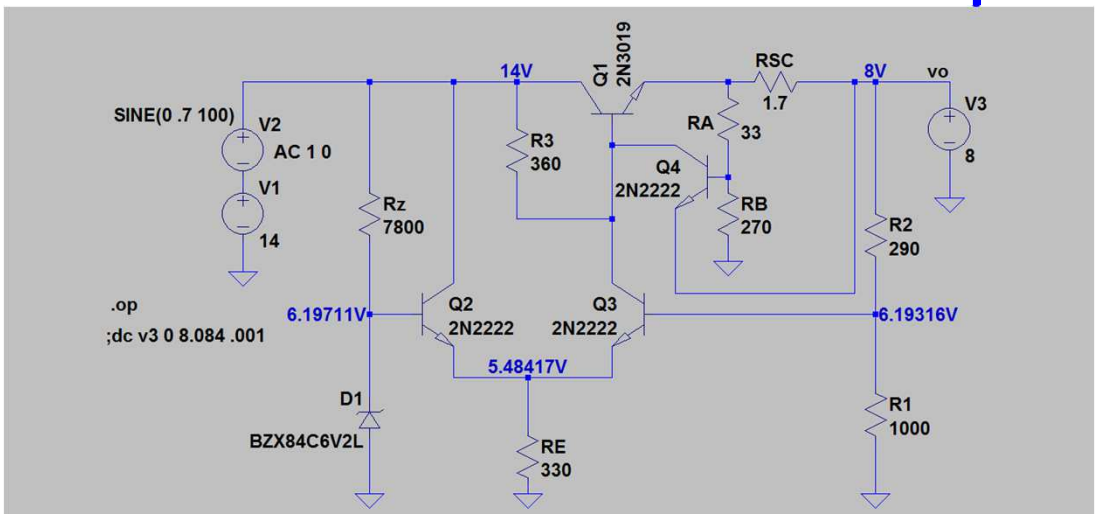
En el repliegue y suponiendo que:

$$I_L = I_{FB} \gg I_{E4} \quad ; \quad I_{B4} \ll I_x$$

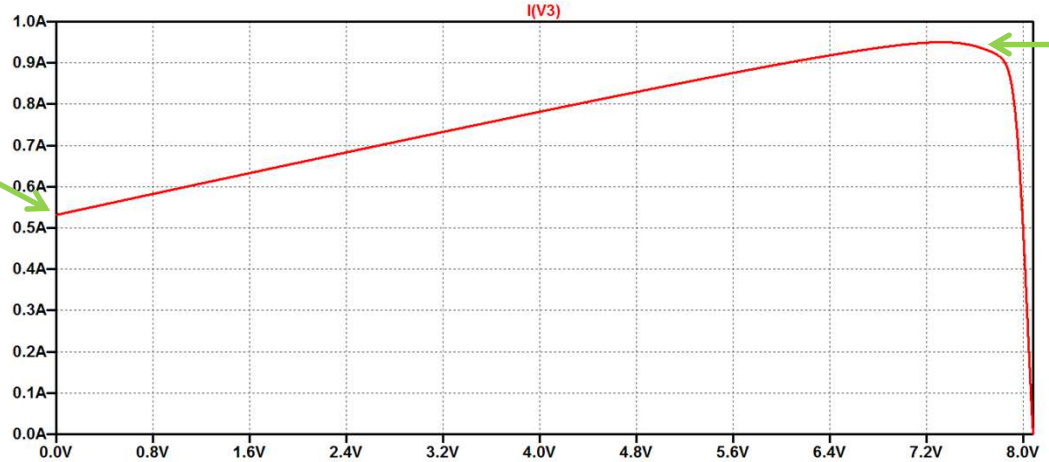
$$I_{FB} \cong \frac{V_{E1}}{R_{SC}} \cong \frac{V_{BE4}}{R_{SC}}\left(\frac{R_A + R_B}{R_B}\right) \qquad (2)$$

$$P_{Q1} = I_{FB}V_i \quad o \quad P_{Q1} = I_{Lmáx}(V_i - V_L)$$

Protección contra CC con repliegue



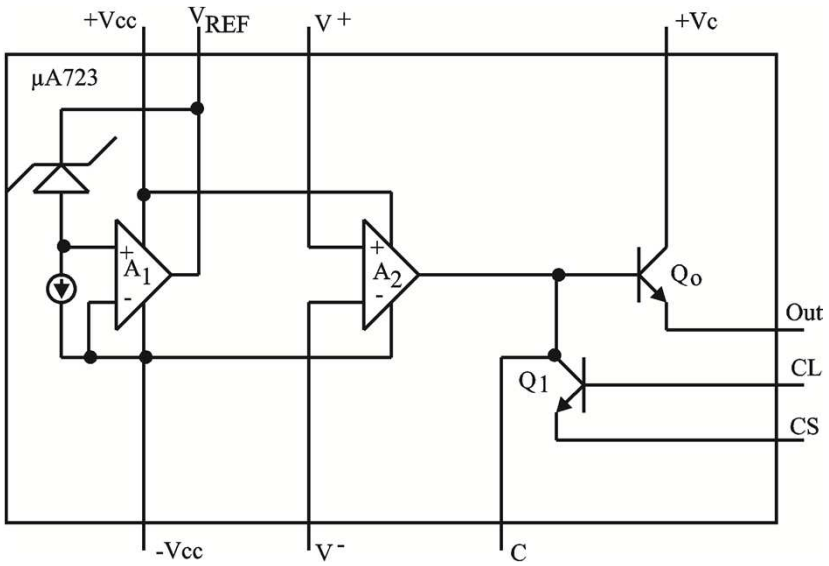
I_{FB}

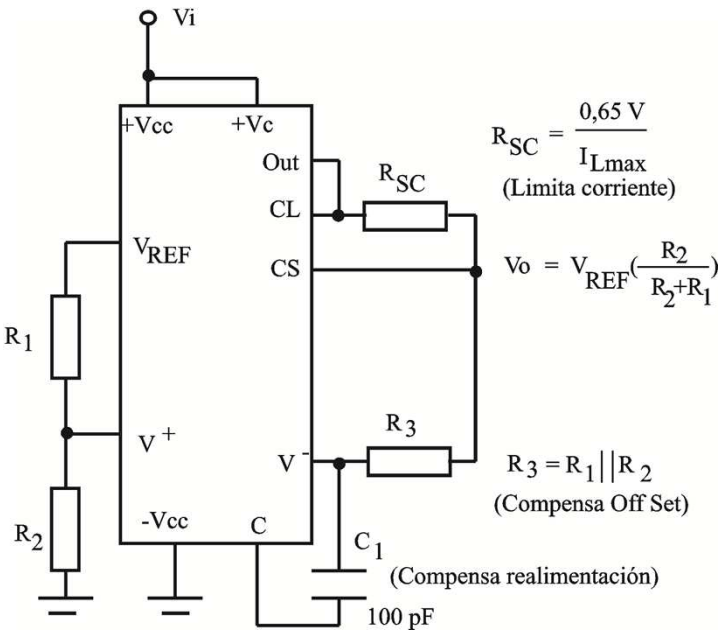


$I_{Lmáx}$

Fuentes reguladas serie integradas

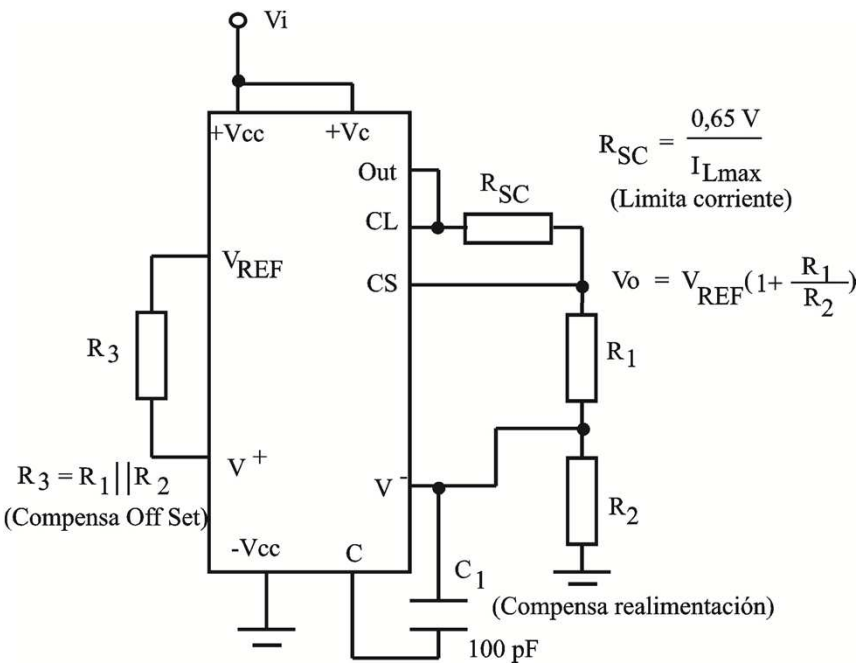
μA723 como regulador entre 2 y 7 V



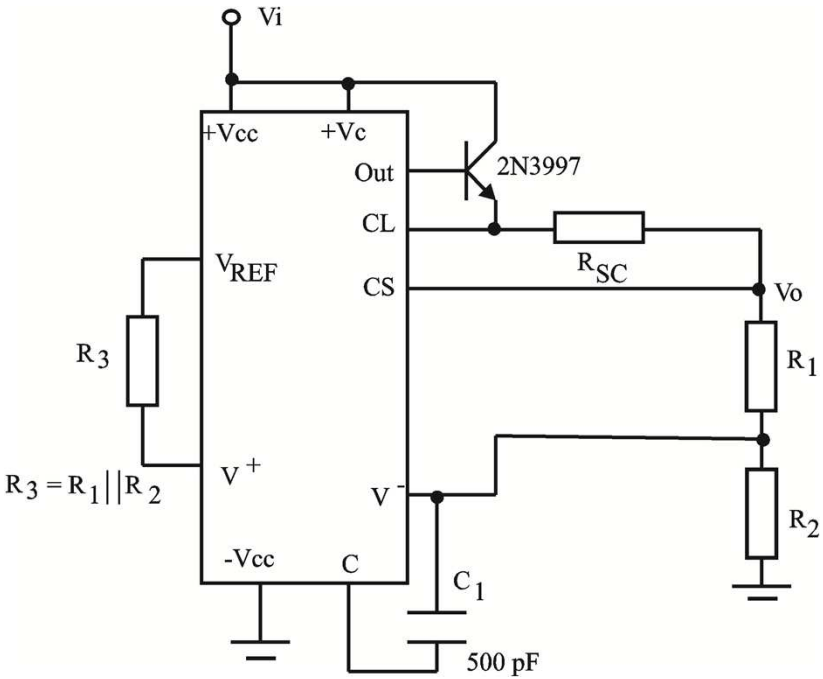


Fuentes reguladas serie integradas

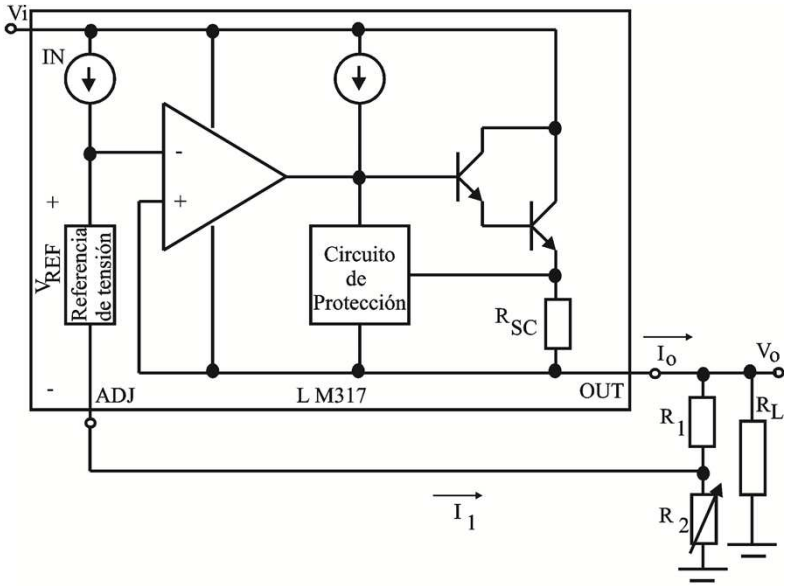
μA723 como regulador entre 7 y 37 V



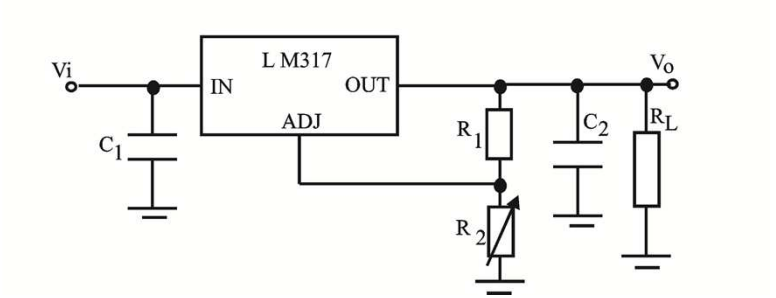
μA723 aumento de la capacidad de corriente



Fuentes reguladas serie integradas



Como I_1 *máxima es* 100 μA :



$$V_o \cong V_{REF}\left(1+\frac{R_2}{R_1}\right)$$